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Similar Judgments, Different Information Processing

Individual Bias in Face Perception and Its Neural Dynamics

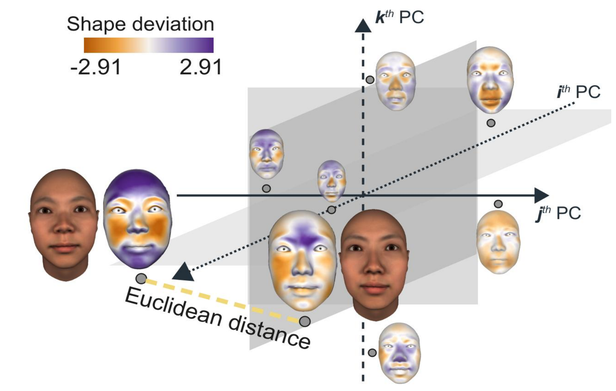
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Introduction

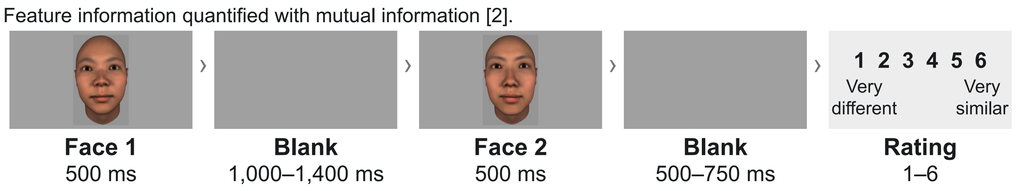
- Similarity judgments support generalization, allowing past experience to guide responses to new encounters [1].
- Yet, similar judgments can arise from distinct feature computations, masking observer-specific processing behind a shared outcome.
- Using 3D faces as a model system, we dissociated individual feature biases from objectively available stimulus information, then tracked the neural dynamics of both with MEG.

Method



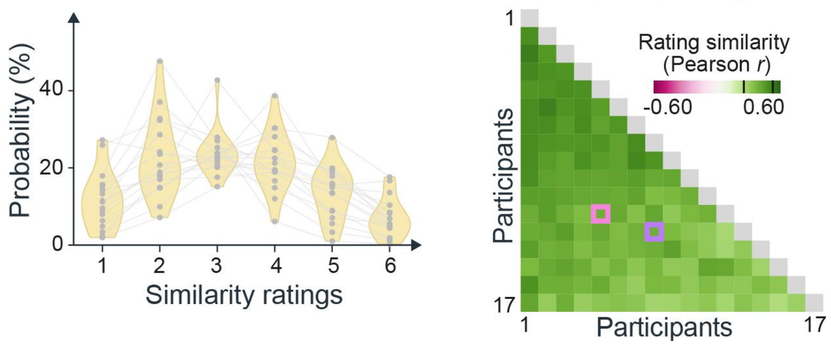
- 17 observers judged the same 720 face pairs.
- We recorded magnetoencephalography (MEG) during sequential presentation.

Feature information quantified with mutual information [2].

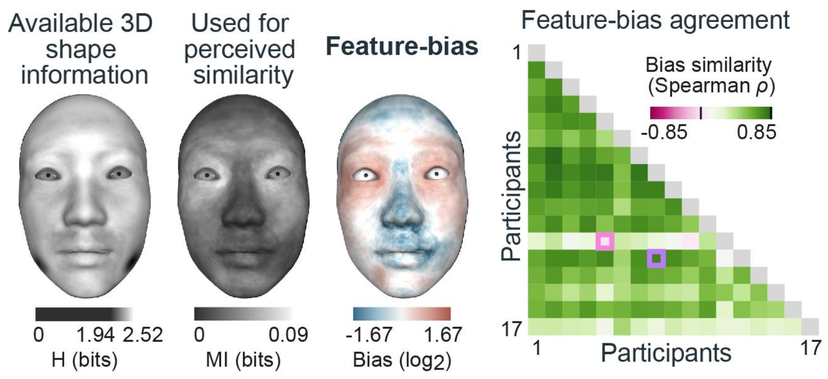


Behavioral Agreement and Idiosyncratic Bias

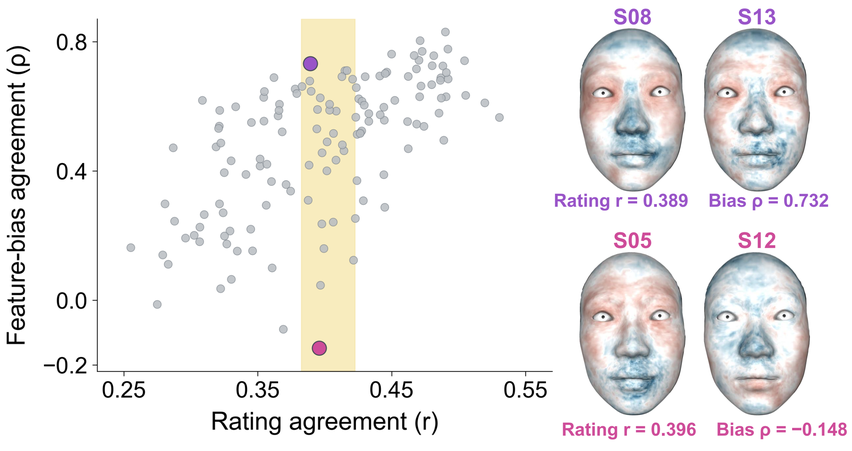
- **Observers broadly agreed on face similarity despite differences in their use of the rating scale**



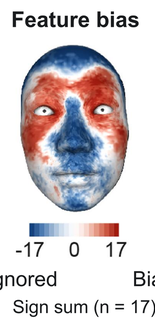
- **Feature bias maps reveal individual differences in the facial information used for similarity judgments**



- **Comparable agreement in similarity ratings can accompany markedly different agreement in feature use**

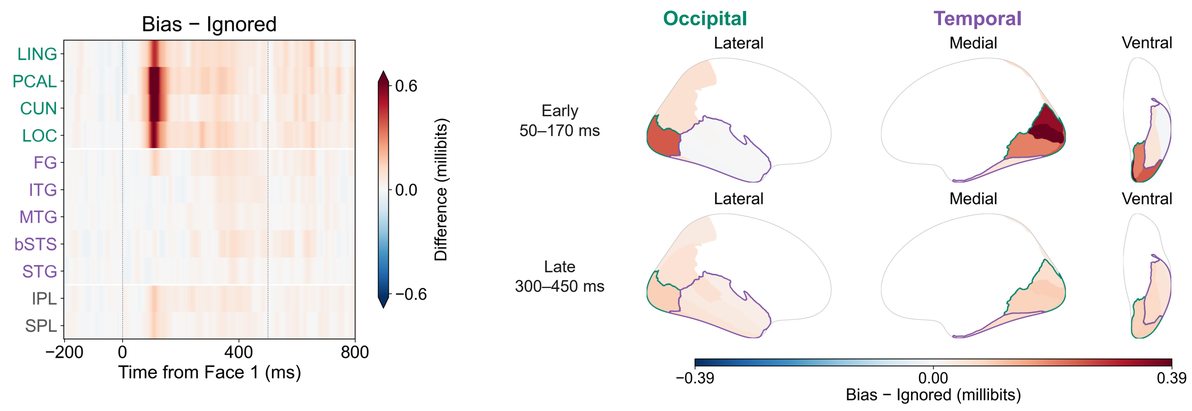


Biased Representation Along Occipital-Ventral regions

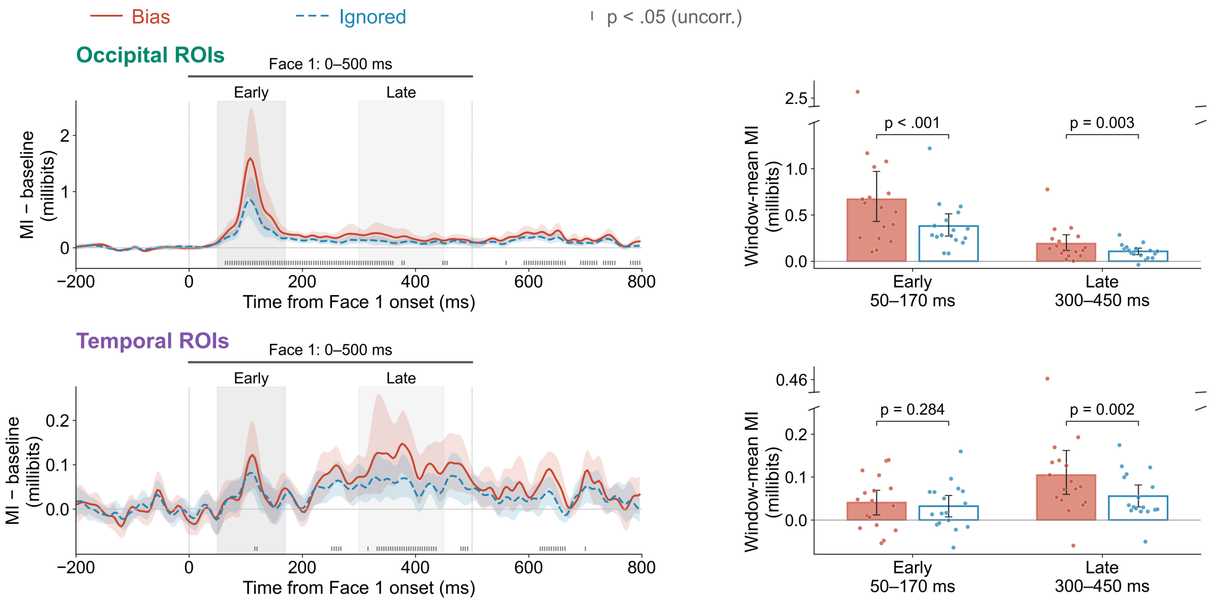


- Bias and Ignored label relatively over- and underused features, defined from each observer's feature bias map.
- Both classes showed an early occipital peak, with stronger encoding of biased features.
- Temporal ROIs showed no clear early class difference, but greater late encoding of biased features.

- **Where and When Feature Bias Is Encoded**

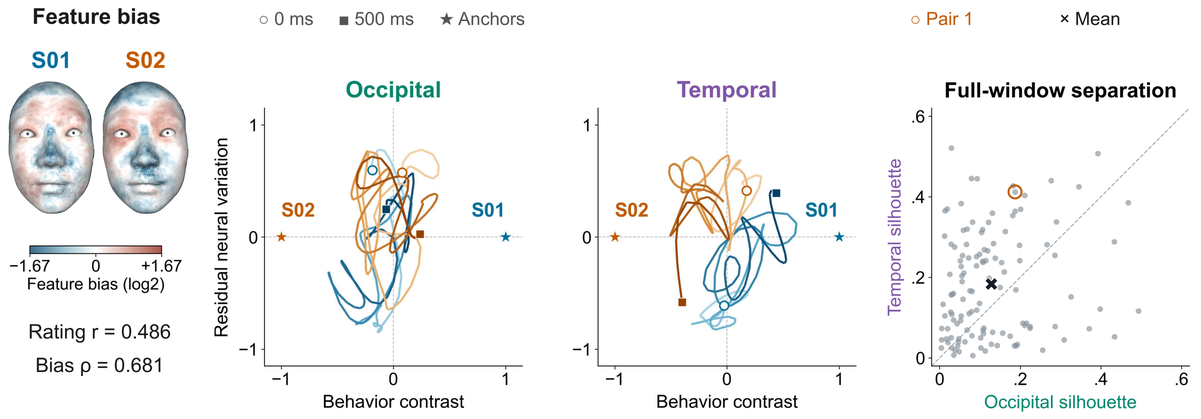


- **Feature Bias Across Early and Late Windows**



Idiosyncratic Neural Trajectories in Temporal Cortex

- **Observer trajectories were more separated in temporal than occipital space**



Conclusion

- Similar judgments can arise from different feature biases across individuals.
- Observer-specific representations emerge later and become increasingly separable in temporal regions, even all stimulus information should be processed.
- **Shared perceptual input – output mappings can arise from idiosyncratic information processing along the occipital-ventral hierarchy.**

[1] Shepard, R. N. (1987). Toward a universal law of generalization for psychological science. Science, 237, 1317–1323.
[2] Zhan et al. (2019). Dynamic Construction of Reduced Representations in the Brain for Perceptual Decision Behavior. Current Biology, 29, 319–326.e4.

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